Outline: Spherical Freezing Analogy for Cosmology

Spherical freezing model

The planar freezing–lake model provides an exact analogue of a radiation–dominated Friedmann equation through the evolution of the ice thickness $s(t)$.[6] Here we show that an analogous result arises in a fully three–dimensional geometry by considering a spherically symmetric freezing problem.

Setup. Consider an ice sphere of radius $R(t)$ growing into an infinite bath of supercooled water. The liquid temperature far from the interface is

$$T(r \to \infty) = T_\infty < T_\phi,$$

where T_ϕ is the equilibrium freezing temperature. The ice–water interface is located at

$$r = R(t),$$

with

$$T(R(t), t) = T_\phi.$$

We assume:

1. spherical symmetry,
2. quasi–static heat conduction,
3. negligible temperature gradient inside the ice.

Temperature field in the liquid. In the freezing problem the temperature field is governed by the heat diffusion equation

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T,$$

where κ is the thermal diffusivity. In typical solidification processes the thermal diffusion time scale is much shorter than the time scale over which the interface position $R(t)$ evolves. Consequently the temperature field relaxes rapidly to a quasi–steady profile compared with the slow motion of the freezing front. Under this quasi–static approximation the time derivative may be neglected, reducing the heat equation to Laplace's equation

$$\nabla^2 T = 0,$$

which determines the instantaneous temperature distribution in each phase. Considering spherical symmetry, in the liquid region $r > R(t)$ the temperature obeys Laplace's equation

$$\nabla^2 T = \frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dT}{dr}\right) = 0.$$

The spherically symmetric solution is

$$T_w(r) = A + \frac{B}{r}.$$

Applying the boundary conditions

$$T_w(R) = T_\phi, \qquad T_w(\infty) = T_\infty,$$

gives

$$T_w(r) = T_\infty + (T_\phi - T_\infty)\frac{R}{r}.$$

The radial gradient is therefore

$$\frac{dT_w}{dr} = -(T_\phi - T_\infty)\frac{R}{r^2}.$$

At the interface

$$\left.\frac{dT_w}{dr}\right|_{R^+} = -\frac{T_\phi - T_\infty}{R}.$$

Stefan condition. The latent heat released during solidification must be conducted away through the liquid. The Stefan condition therefore gives

$$\rho_i L\,\dot{R} = -\,\lambda_w \left.\frac{dT_w}{dr}\right|_{R^+},$$

where ρ_i is the ice density, L the latent heat of fusion, and λ_w the thermal conductivity of water.

Substituting the interface gradient yields

$$\rho_i L\,\dot{R} = \lambda_w \frac{T_\phi - T_\infty}{R}.$$

Defining the constant

$$\alpha \equiv \frac{\lambda_w (T_\phi - T_\infty)}{\rho_i L},$$

we obtain the spherical Stefan growth law

$$\dot{R} = \frac{\alpha}{R}. \tag{1}$$

Exact solution. Multiplying Eq. (1) by R gives

$$R\dot{R} = \alpha.$$

Hence

$$\frac{d}{dt}(R^2) = 2\alpha,$$

so that

$$R^2(t) = R_0^2 + 2\lambda(t - t_0). \tag{2}$$

At late times the growth law becomes

$$R(t) \propto t^{1/2}.$$

Friedmann analogue. Define the spherical Hubble parameter

$$H_R \equiv \frac{\dot{R}}{R}.$$

Using Eq. (1) we obtain

$$H_R = \frac{\alpha}{R^2}, \qquad H_R^2 = \alpha^2 R^{-4}. \tag{3}$$

This has exactly the form of a Friedmann equation for a spatially flat universe containing radiation,

$$H^2 \propto a^{-4}.$$

With the identification

$$a(t) \longleftrightarrow R(t),$$

the spherical freezing problem therefore reproduces the radiation–era expansion law

$$a(t) \propto t^{1/2}.$$

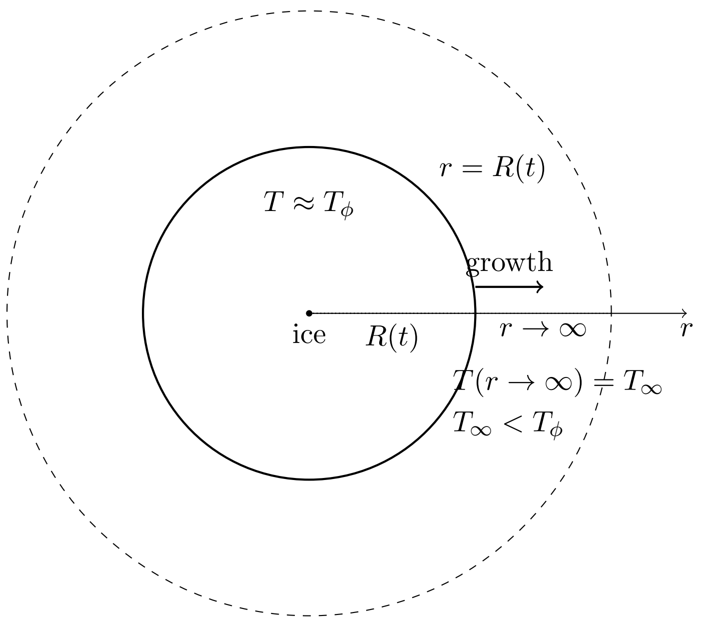

Figure 1: Spherical freezing model

Effective energy density and equation of state. Writing the evolution equation as

$$H_R^2 = \frac{8\pi G}{3}\rho_{\text{eff}}(R),$$

one finds

$$\rho_{\text{eff}}(R) = \frac{3\alpha^2}{8\pi G} R^{-4}.$$

Comparing with

$$\rho(a) \propto a^{-3(1+w)},$$

gives $w = \frac{1}{3}$.

Thus the spherical freezing model with an infinite thermal reservoir provides an *exact* analogue of a radiation–dominated Friedmann universe. This result parallels the planar freezing–lake model, but arises from the geometric spreading of radial heat flux rather than from the growth of a planar conductive layer.

Spherical freezing with an actively cooled isothermal core

The previous spherical model assumes an ice sphere growing into supercooled water, $T_\infty < T_\phi$, so that the latent heat released at the interface is conducted away through the liquid. This gives a one-sided Stefan condition and leads directly to the radiation-like scaling $H_R^2 \propto R^{-4}$.

A more experimentally controllable realization is to place a small actively cooled core at the center of the growing ice sphere. The core is maintained at a fixed temperature

$$T_0 < T_\phi$$

by an external cooling mechanism. The surrounding water bath may now be warmer than the freezing point,

$$T_\infty > T_\phi,$$

so that the liquid tends to melt the interface, while the cooled core removes heat and drives freezing.

Setup. Consider a small spherical core of radius a, with

$$a \ll R(t),$$

placed at the center of an ice shell of outer radius $R(t)$. The core is assumed to be thermostatted at a fixed temperature $T_0 < T_\phi$. Physically this may be realized by a high-conductivity metallic sphere or a small heat exchanger connected to an external cooling bath. The ice–water interface remains at

$$T(R(t), t) = T_\phi.$$

We assume:

1. spherical symmetry,
2. quasi-static heat transport,
3. an effectively isothermal cooled core at temperature T_0,
4. a reduced thermal-resistance description of the ice-side heat flux.

Effective ice-side heat flux. Instead of resolving the detailed heat-transfer mechanism inside the cooled core, we model the core and its cooling apparatus as an effective heat sink, Thus we write

$$q_i^{\text{eff}} = \lambda_i \frac{T_\phi - T_0}{R}, \tag{4}$$

where λ_i is the thermal conductivity of ice.

Water-side heat flux. In the water region, assuming spherical symmetry and quasi-static heat conduction, the temperature field satisfies

$$\nabla^2 T_w = 0,$$

with boundary conditions

$$T_w(R) = T_\phi, \qquad T_w(\infty) = T_\infty.$$

Hence

$$T_w(r) = T_\infty + (T_\phi - T_\infty)\frac{R}{r}.$$

The radial derivative at the interface is

$$\left.\frac{dT_w}{dr}\right|_{R^+} = -\frac{T_\phi - T_\infty}{R}.$$

Therefore the water-side contribution to the Stefan balance is

$$q_w = -\lambda_w \left.\frac{dT_w}{dr}\right|_{R^+} = \lambda_w \frac{T_\phi - T_\infty}{R}. \tag{5}$$

If the bath is warmer than the freezing point, $T_\infty > T_\phi$, then this term is negative and opposes freezing.

Two-sided effective Stefan balance. The latent heat released at the moving interface is balanced by the net heat flux away from the interface:

$$\rho_i L \dot{R} = q_i^{\text{eff}} + q_w. \tag{6}$$

Using Eqs. (4) and (5), we obtain

$$\rho_i L \dot{R} = \lambda_i \frac{T_\phi - T_0}{R} + \lambda_w \frac{T_\phi - T_\infty}{R}. \tag{7}$$

Define

$$\alpha_{\text{core}} \equiv \frac{\lambda_i (T_\phi - T_0) + \lambda_w (T_\phi - T_\infty)}{\rho_i L}. \tag{8}$$

Then Eq. (7) becomes

$$\dot{R} = \frac{\alpha_{\text{core}}}{R}. \tag{9}$$

Growth condition. Freezing requires

$$\alpha_{\text{core}} > 0.$$

If $T_\infty > T_\phi$, this means

$$\lambda_i (T_\phi - T_0) > \lambda_w (T_\infty - T_\phi). \tag{10}$$

Thus the cooled core must remove heat faster than the warm water bath supplies it to the interface.

Friedmann analogue. Equation (9) has the same mathematical form as the supercooled-bath Stefan law. Therefore

$$R\dot{R} = \alpha_{\text{core}},$$

and

$$R^2(t) = R_0^2 + 2\alpha_{\text{core}}(t - t_0). \tag{11}$$

The spherical Hubble parameter is

$$H_R = \frac{\dot{R}}{R} = \frac{\alpha_{\text{core}}}{R^2},$$

so that

$$H_R^2 = \alpha_{\text{core}}^2 R^{-4}. \tag{12}$$

Thus the actively cooled-core model also realizes the radiation-like Friedmann scaling

$$H^2 \propto a^{-4},$$

with the identification

$$a(t) \longleftrightarrow R(t).$$

Experimental realization. A possible laboratory setup consists of a small spherical metallic core or heat exchanger placed at the center of a transparent water tank. The core is connected to an external cooling loop which maintains it at a fixed temperature $T_0 < T_\phi$. A thin initial ice layer is first formed around the core, and the subsequent growth of the ice shell is monitored optically.

The water bath temperature T_∞ can be chosen above T_ϕ to study freezing driven primarily by heat extraction through the cooled core, testing the condition

$$\lambda_i(T_\phi - T_0) > \lambda_w(T_\infty - T_\phi).$$

The main observables are:

$$R(t), \qquad \dot{R}(t), \qquad H_R(t) = \frac{\dot{R}}{R}.$$

The key scaling test is

$$R^2(t) \simeq R_0^2 + 2\alpha_{\text{core}} t, \tag{13}$$

or equivalently

$$H_R^2 R^4 \simeq \alpha_{\text{core}}^2 = \text{constant}. \tag{14}$$

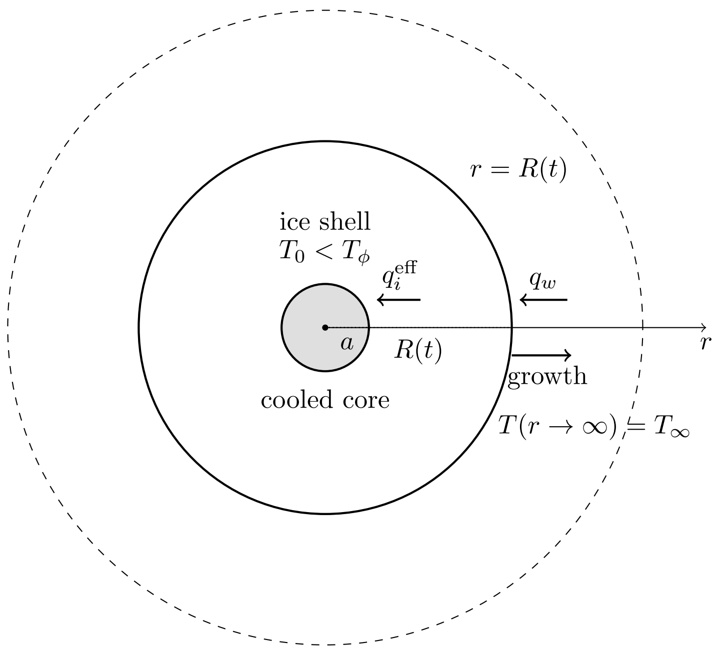

Figure 2: Actively cooled isothermal-core model.

Numerical simulation The evolution of the ice radius in the model is compared with that of the scale factor in a simple CDM universe in Fig 3, constants are provided in Table 1

Actively cooled spherical freezing with buoyancy-driven convection

We now combine the actively cooled-core model with buoyancy-driven heat transport in the surrounding liquid. The cooled core removes heat through the ice shell, while the water-side flux is enhanced and distorted by convection. Gravity selects a preferred vertical axis, so the interface is naturally described as a weakly deformed axisymmetric sphere.

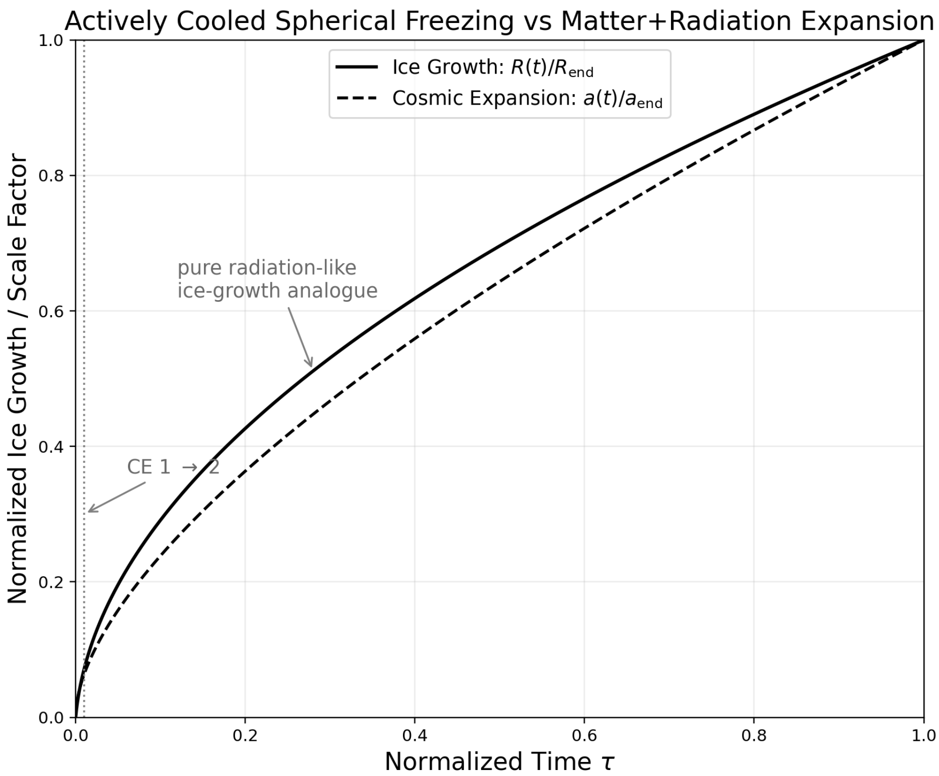

Figure 3: Normalized ice thickness $R(t)/R_{\text{end}}$ in the spherical freezing with an actively cooled isothermal core(solid curve) compared with the normalized scale factor $a(t)/a_0$ in a matter–radiation cosmological expansion (CE) model with zero cosmological constant ($\Lambda = 0$), both plotted as functions of normalized time τ.

Axisymmetric interface. We parametrize the ice–water interface by

$$R(\theta, t) = S(t)\left[1 + \varepsilon(t) P_2(\cos\theta)\right], \qquad |\varepsilon| \ll 1, \tag{15}$$

where $S(t)$ is the mean radius, $\varepsilon(t)$ is the quadrupolar anisotropy, and

$$P_2(\mu) = \frac{1}{2}(3\mu^2 - 1). \tag{16}$$

The polar and equatorial radii are therefore, to linear order,

$$R_{\parallel} = S(1 + \varepsilon), \qquad R_{\perp} = S\left(1 - \frac{\varepsilon}{2}\right). \tag{17}$$

This is the same kinematical structure as a locally rotationally symmetric Bianchi-I geometry, where one direction expands differently from the two transverse directions [2].

Effective Stefan condition. The local Stefan balance is taken to be

$$\rho_i L\, \partial_t R(\theta, t) = \lambda_i \frac{T_\phi - T_0}{R(\theta, t)} + \lambda_w \frac{T_\phi - T_\infty}{R(\theta, t)} Nu(\theta, t), \tag{18}$$

where $T_0 < T_\phi$ is the actively maintained core temperature and $Nu(\theta, t)$ is the local water-side Nusselt factor. The first term describes effective heat extraction by the cooled core; the second describes heat exchange with the liquid. If $T_\infty > T_\phi$, then the water-side term is negative and opposes freezing.

Define

$$A_i = \frac{\lambda_i(T_\phi - T_0)}{\rho_i L}, \qquad A_w = \frac{\lambda_w(T_\phi - T_\infty)}{\rho_i L}. \tag{19}$$

Then

$$\partial_t R(\theta, t) = \frac{A_i + A_w Nu(\theta, t)}{R(\theta, t)}. \tag{20}$$

We model the convective anisotropy by

$$Nu(\theta, t) = \bar{N}u(S)\left[1 + \xi(S) P_2(\cos\theta)\right]. \tag{21}$$

This reduced Nusselt description is not meant to resolve the full hydrodynamic flow; it encodes the net effect of water-side transport on the moving interface. Such Nusselt–Rayleigh closures are standard in buoyancy-driven heat transfer, where one often writes $Nu \sim Ra^\beta$ over a given scaling regime [1]. The anisotropic part, controlled by $\xi(S)$, plays the role of a small effective anisotropic stress source, analogous to the skew-stress sources studied in anisotropic cosmology [3, 4, 5].

Introducing

$$\Lambda(S) = A_i + A_w \bar{N}u(S), \qquad \mathcal{X}(S) = A_w \bar{N}u(S)\xi(S), \tag{22}$$

Eq. (20) becomes

$$\partial_t R(\theta, t) = \frac{\Lambda(S) + \mathcal{X}(S) P_2(\cos\theta)}{R(\theta, t)}. \tag{23}$$

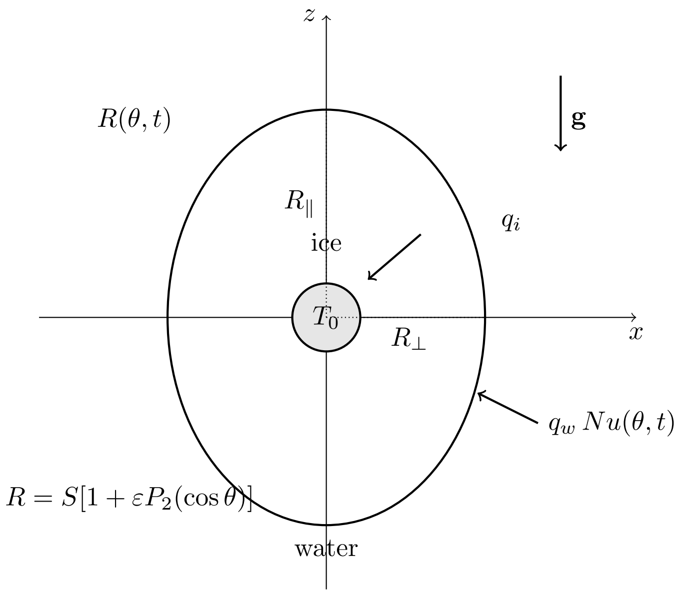

Figure 4: Actively cooled spherical freezing with anisotropic water-side convection.

Reduced evolution equations. Substituting Eq. (15) into Eq. (23) and expanding to first order in ε gives

$$\partial_t R = \dot{S} + \left(\dot{S}\,\varepsilon + S\dot{\varepsilon}\right) P_2, \tag{24}$$

while

$$\frac{\Lambda + \mathcal{X} P_2}{R} \simeq \frac{\Lambda}{S} + \frac{\mathcal{X} - \Lambda\varepsilon}{S} P_2. \tag{25}$$

Equating the isotropic and quadrupolar parts yields

$$\dot{S} = \frac{\Lambda(S)}{S}, \tag{26}$$

and

$$\dot{\varepsilon} = \frac{1}{S^2}\left[\mathcal{X}(S) - 2\Lambda(S)\varepsilon\right]. \tag{27}$$

Equivalently,

$$\dot{\varepsilon} = \frac{\Lambda(S)}{S^2}\left[\xi_{\text{eff}}(S) - 2\varepsilon\right], \tag{28}$$

where

$$\xi_{\text{eff}}(S) = \frac{\mathcal{X}(S)}{\Lambda(S)} = \frac{A_w \bar{N}u(S)}{A_i + A_w \bar{N}u(S)}\,\xi(S). \tag{29}$$

Thus the isotropic cooled-core flux screens the anisotropic forcing generated on the water side.

Mean Friedmann analogue. Assume the reduced convective scaling

$$Ra \propto S^3, \qquad \bar{N}u(S) = Nu_0 S^{3\beta}, \qquad B \equiv A_w Nu_0. \tag{30}$$

Then

$$\Lambda(S) = A_i + BS^{3\beta}, \tag{31}$$

and

$$H \equiv \frac{\dot{S}}{S} = \left(A_i + BS^{3\beta}\right) S^{-2}. \tag{32}$$

Therefore

$$H^2 = A_i^2 S^{-4} + 2A_i B\, S^{3\beta-4} + B^2 S^{6\beta-4}. \tag{33}$$

The cooled core gives the radiation-like term S^{-4}, while water-side convection generates the additional mixed and convective contributions. For $\beta = 1/3$,

$$H^2 = A_i^2 S^{-4} + 2A_i B\, S^{-3} + B^2 S^{-2}, \tag{34}$$

which has the formal structure of radiation-, matter-, and curvature-like terms. This identification should be understood as a structural analogy between evolution equations, not as a literal gravitational dynamics.

Anisotropic expansion and shear. Define directional scale factors

$$a_{\parallel} = S(1+\varepsilon), \qquad a_{\perp} = S\left(1 - \frac{\varepsilon}{2}\right), \tag{35}$$

so that $a_{\parallel} a_{\perp}^2 \simeq S^3$. Their Hubble rates are

$$H_{\parallel} \simeq H + \dot{\varepsilon}, \qquad H_{\perp} \simeq H - \frac{1}{2}\dot{\varepsilon}. \tag{36}$$

The LRS shear scalar is

$$\sigma^2 = \frac{1}{3}(H_{\parallel} - H_{\perp})^2 = \frac{3}{4}\dot{\varepsilon}^2. \tag{37}$$

This is the same shear construction used in LRS Bianchi-I cosmology [2].

If the anisotropic forcing decays, $\xi_{\text{eff}} \to 0$, then Eq. (27) gives

$$\frac{d\varepsilon}{dS} = -\frac{2\varepsilon}{S}, \tag{38}$$

and hence

$$\varepsilon = \varepsilon_0 S^{-2}. \tag{39}$$

Using Eq. (26),

$$\dot{\varepsilon} = -2\varepsilon_0 \left(A_i + BS^{3\beta}\right) S^{-4}, \tag{40}$$

so

$$\sigma^2 = 3\varepsilon_0^2 \left(A_i + BS^{3\beta}\right)^2 S^{-8}. \tag{41}$$

The conservative anisotropic Friedmann form is therefore

$$H^2 = \left(A_i + BS^{3\beta}\right)^2 S^{-4} + \varepsilon_0^2 \left(A_i + BS^{3\beta}\right)^2 S^{-8}. \tag{42}$$

The first term represents the mean expansion analogue, while the second term plays the role of the shear contribution. In the core-dominated regime one recovers

$$H^2 \propto S^{-4}, \qquad \sigma^2 \propto S^{-8}.$$

In the convection-dominated regime,

$$H^2 \propto S^{6\beta-4}, \qquad \sigma^2 \propto S^{6\beta-8}.$$

For $\beta = 1/3$, the shear term scales as S^{-6}, matching the standard LRS Bianchi-I shear scaling [2]. More generally, when ξ_{eff} remains nonzero, the model is better interpreted as a sourced anisotropic expansion, analogous to Bianchi-I cosmologies with anisotropic stresses rather than a freely decaying shear sector [3, 4, 5].

Next step: Experiment/Simulation

A Simulation Constants

Symbol	Value	Unit	Description / Source
ρ_i	917	$\mathrm{kg\,m^{-3}}$	Density of ice near 0°C.
L	3.34×10^5	$\mathrm{J\,kg^{-1}}$	Latent heat of fusion of ice
λ_i	2.22	$\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of ice
λ_w	0.56	$\mathrm{W\,m^{-1}\,K^{-1}}$	Thermal conductivity of water
T_ϕ	0.0	°C	Ice–water interface temperature.
T_∞	3.0	°C	Bulk water-bath temperature.
T_0	-20.0	°C	Maintained temperature of the cooled core.
a	2.0×10^{-3}	m	Radius of the actively cooled core.
h_w	5.0	$\mathrm{W\,m^{-2}\,K^{-1}}$	water-side convective heat-transfer coefficient.

Table 1: Physical constants and model parameters used in the numerical simulations of the actively cooled spherical freezing model.

References

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